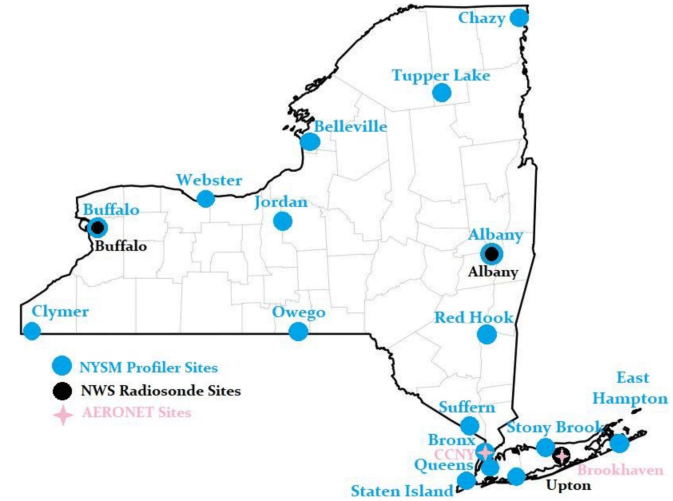


RF Interference in the NYS Mesonet Weather Radiometers

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12/05/2025

Background

- 17 microwave radiometers monitor 22-30 GHz and 51-59 GHz atmospheric radiation across New York State
- From 2018 to 2022, there have been more instances of radiometer brightness temperature exceeding 350 K or dipping below 0 K, which indicate radio frequency interference
- The possibly inaccurate brightness temperature data could be disrupting temperature, relative humidity, water vapor density and liquid density predictions

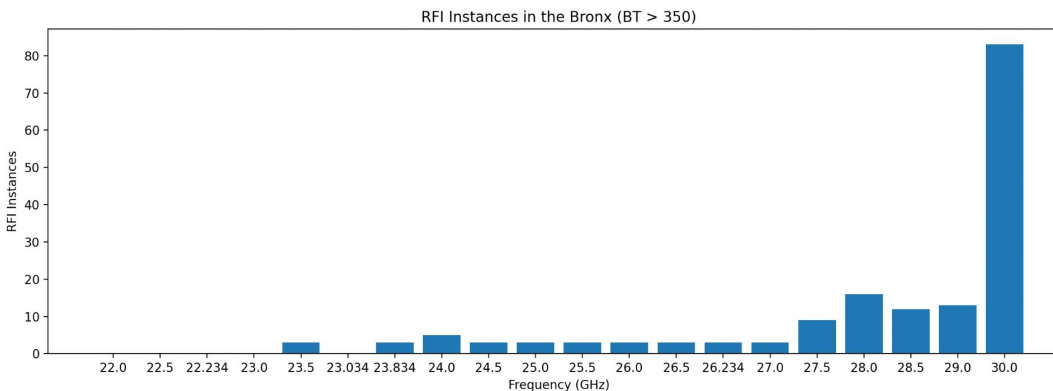
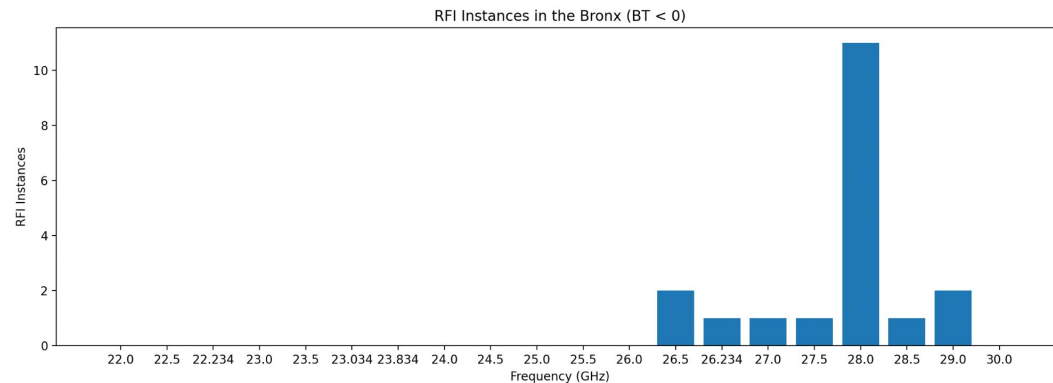


Goals

- Display RF interference in a revealing way to better understand trends across years and frequencies.
- Create a new neural network resistant to RF interference, allowing for more accurate water vapor density predictions

Instances of RF Interference

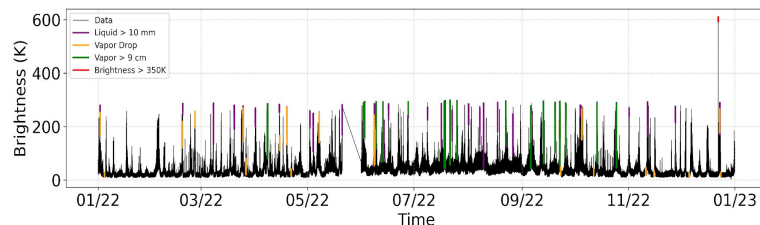
- Data was collected across 2022, 2023, and 2024
- For each frequency, the instances where the brightness temperature exceeded 350 K or dipped below 0 K were summed up and plotted



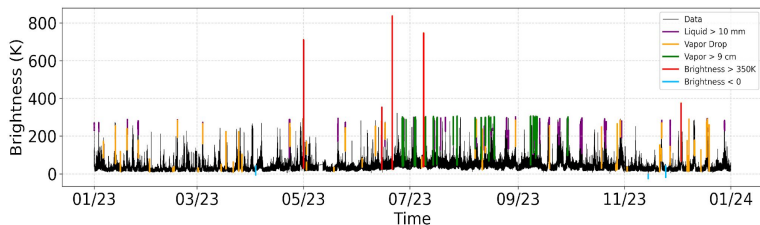
Instances of RF Interference

Bronx 28 GHz

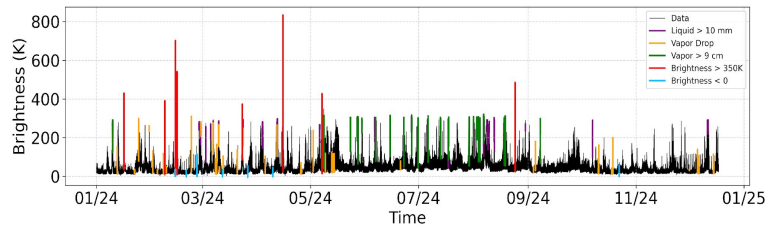
2022



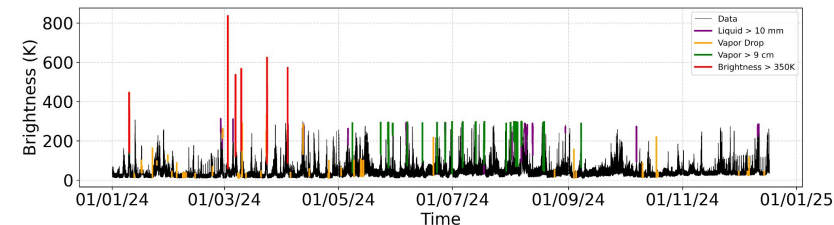
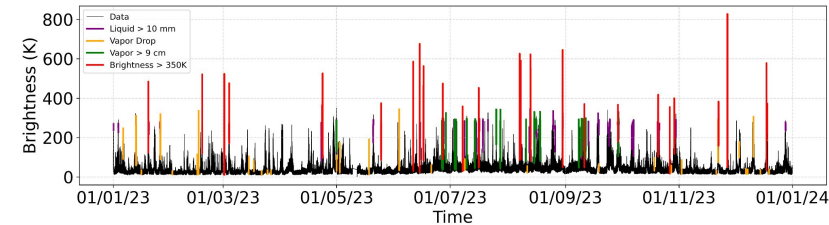
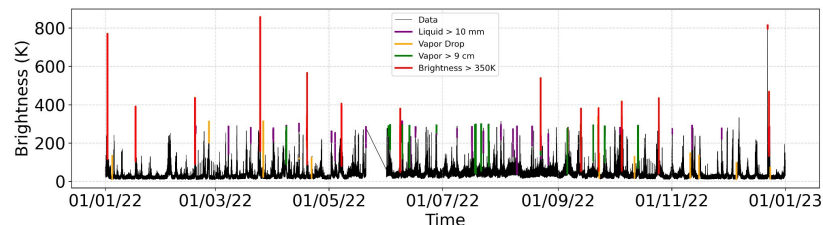
2023



2024



Bronx 30 GHz



Reconsidering Instances of RFI

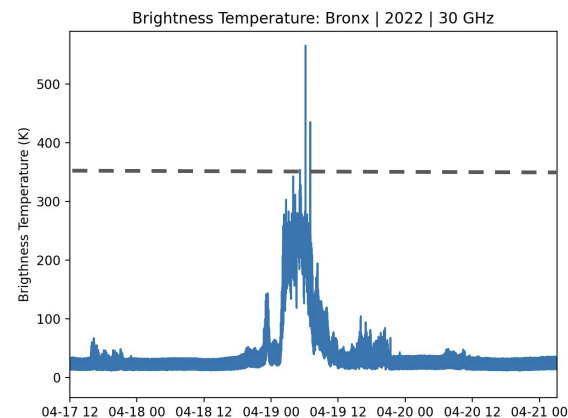
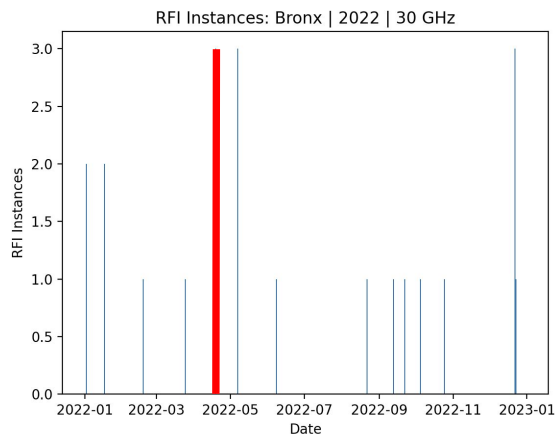
Some interference counts were unreasonable

Problem

- A single interference event occasionally lasts long enough to produce more than one RFI data point

Solution

- If a certain day has multiple instances of RFI, consider all of that as one RFI count. The total number of days with RFI present is the total RFI count.



The red bar indicates three instances of RFI on April 19, 2022. Correspondingly, The dashed line indicates 3 instances of brightness temperature above 350 K. Those 3 data points lying above 350 K seem to belong to the same RFI event

Reconsidering Instances of RFI

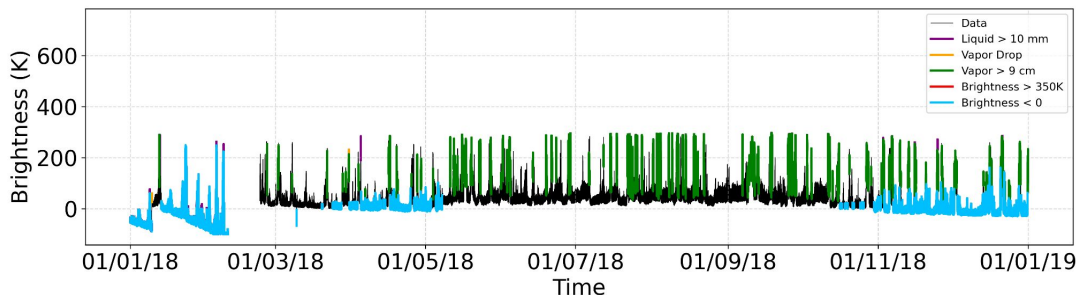
Some interference counts were unreasonable

Problem

- Many time the brightness temperature dipped below 0 for long periods of time

Solution

- Only consider instances where the brightness temperature exceeds 350 K as RFI

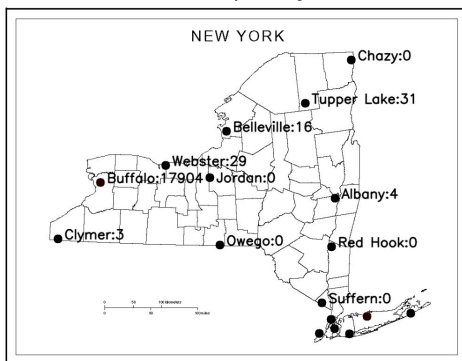


Displaying the RF Interference Across New York State

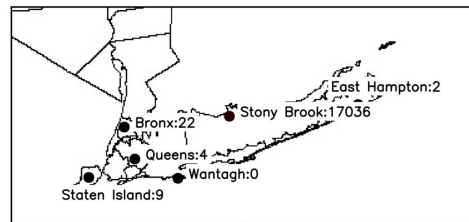
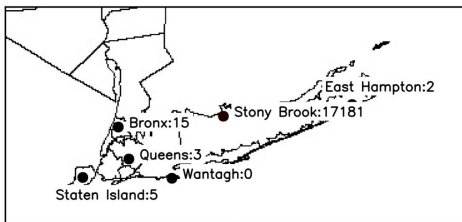
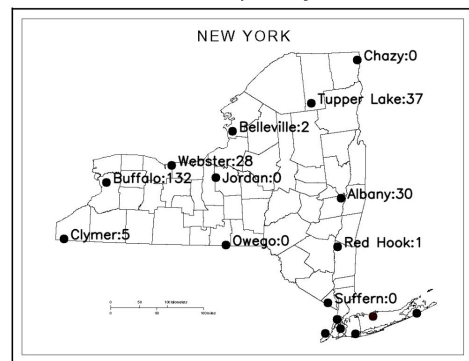
- Each radiometer location is indicated on the map with a dot
- The dot color ranges from black (low RFI) to red (high RFI)
- The number of RFI instances is displayed next to each radiometer name
- **A map for each year and frequency was created**

*An RFI instance counts as a data point that exceeds 350K or goes below 0K

Year: 2024 Frequency: 28.0 GHz

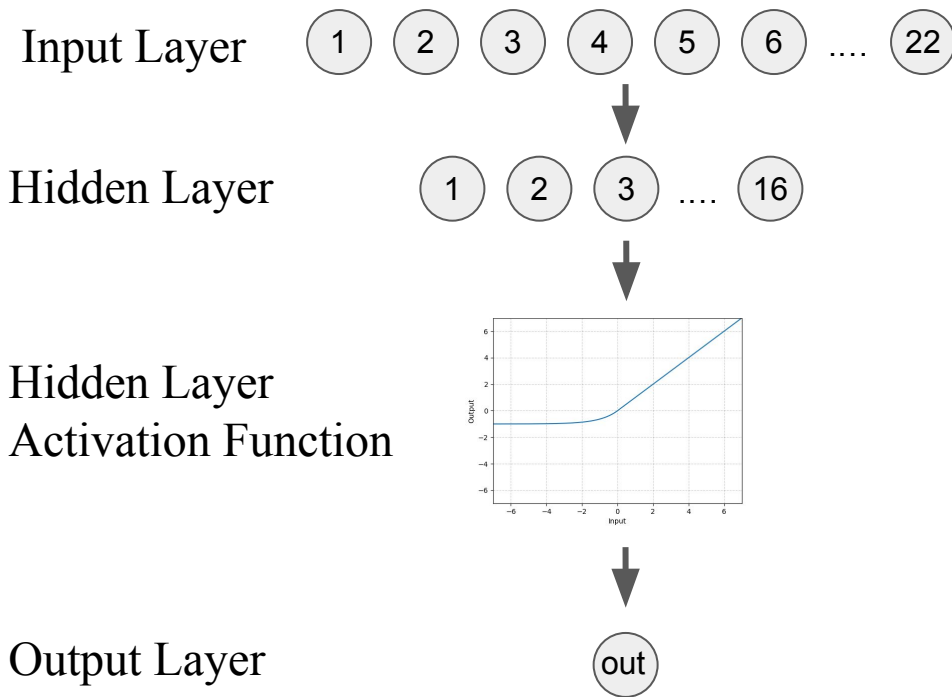


Year: 2024 Frequency: 30.0 GHz



Vapor Neural Network Structure

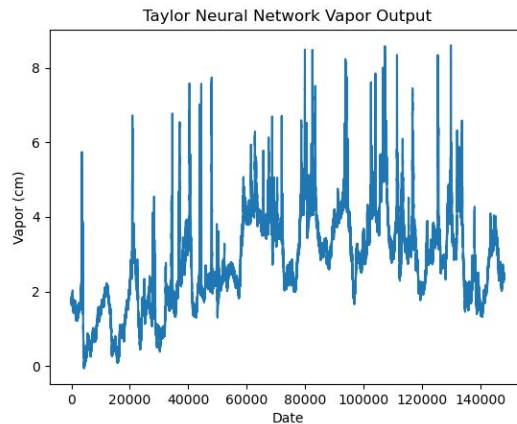
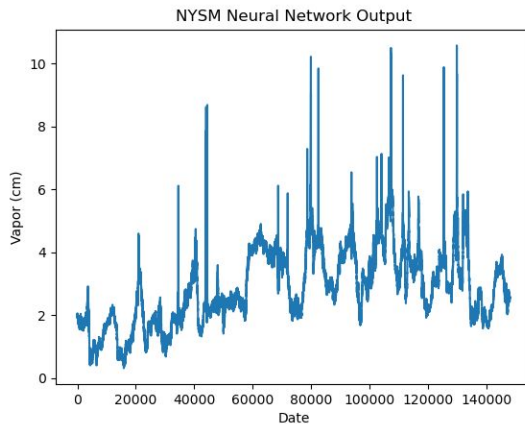
- 22 inputs corresponding to the 22 frequency bins
- 16 hidden neurons
- Hidden neuron activation function
- 1 output neuron



Creating the Neural Network

Neural Network Parameters

- Number of hidden neurons: 16
- Hidden layer activation function: ELU
- Output layer activation function: None
- Criterion: MSELoss
- Optimizer: Adam
- Learning Rate: 0.001
- Weight Decay: 0
- Epochs: 100
- Smoothing window: 25



Training/Testing Parameters

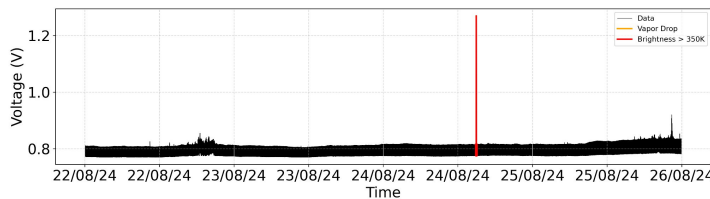
- Training data: Bronx 2022 w/ interference and Bronx 2024 w/ interference
- Test data: Bronx 2023 w/ interference

Trail	Mean Absolute Error
1	0.20591362
2	0.23862307
3	0.312723

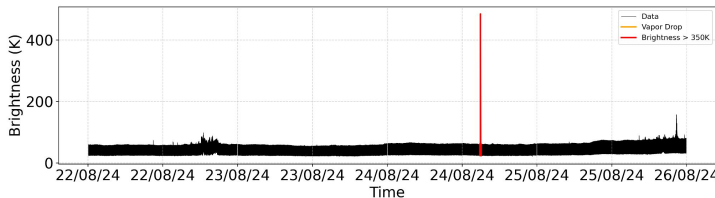
Testing the Neural Network

Bronx | Aug 2024 | 28.0 GHz

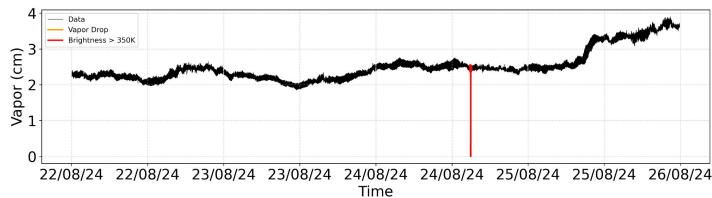
Voltage



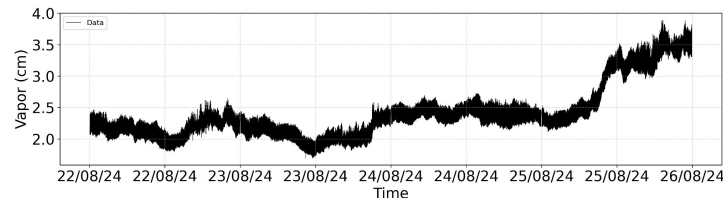
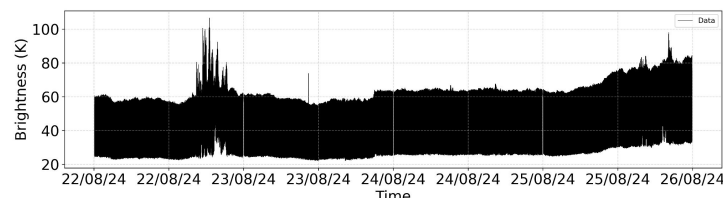
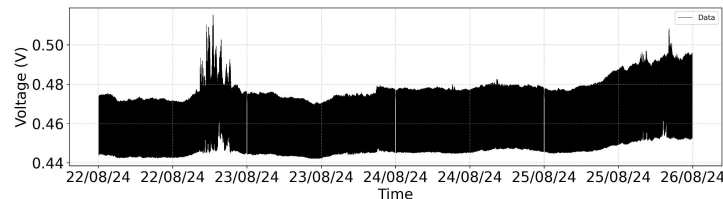
Brightness
Temperature



Water Vapor
Density



Queens | Aug 2024 | 28.0 GHz



Next steps

- Finish calculating the RFI Instances with the binning method: If a certain day has multiple instances of RFI, consider all of that as one RFI count. The total number of days with RFI present is the total RFI count
- Gain a better understanding of what counts as RFI
- Neural network must scale without references to the NYSM Mesonet neural network
- Make the neural network less sensitive to RFI interference